

بسم الله الرحمن الرحيم



Faculty of medicine

Medical Microbiology and Immunology

Department

A thesis submitted in partial fulfillment of the requirements for the Medical master degree of
Medical Microbiology and Immunology

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2025

Introduction

- **The genus *Candida* is a normal component of the human microbiota** [1]. Under certain pathological conditions, however, these yeasts may shift from harmless colonizers to pathogens that cause either **superficial infections** of the skin and mucous membranes or **systemic, invasive disease**.
- *Candida* species rank as the **fourth most common cause of nosocomial bloodstream infections** [2, 3]. Among them, ***Candida albicans* (C. albicans)** remains the species most frequently isolated from bloodstream infections [4], although ***C. glabrata* and *C. tropicalis*** are also prominent pathogens.
- **Predisposing factors** for invasive candidiasis include a variety of **immunocompromised states** (e.g., neutropenia, chemotherapy, organ transplantation) [5]. Notably, **early laboratory diagnosis** can reduce the mortality of invasive *Candida* infections from ~40 % to ~15 % when appropriate antifungal therapy is initiated promptly [7].

- The pathogenic success of *C. albicans* is linked to several **virulence attributes** [8]:
 - Transition from **budding yeast to filamentous hyphae**
 - Expression of **adhesins**
 - **Biofilm formation**
 - Secretion of **hydrolytic enzymes**
- Within this arsenal, **Hyphal Wall Protein 1 (Hwp1)** plays a key role in **adhesion and biofilm formation** [9], while the **Agglutinin-Like Sequence 1 gene (ALS1)** mediates attachment specifically to **endothelial cells** [10].
- Although *C. albicans* remains a leading cause of **candidemia**, many countries report a continuing **shift toward non-albicans *Candida* (NAC) species**. Collectively, *C. albicans*, *C. tropicalis*, *C. parapsilosis*, *C. krusei*, and *C. glabrata* now account for **>90 % of candidemia cases**, posing a growing threat to hospitalized patients [3].

Background

- In *Candida albicans*, **adhesion** and **biofilm formation** are primarily mediated by **Hyphal Wall Protein 1 (Hwp1)**.
- Additionally, adhesion is facilitated by the **Agglutinin-Like Sequence 1 gene (ALS1)**, which plays a crucial role in the attachment of *C. albicans* to **endothelial cells**.
- Although *C. albicans* remains a major cause of **candidemia**, a **global shift** towards **non-*albicans* *Candida* (NAC) species** has been increasingly reported.
- Species such as *C. tropicalis*, *C. parapsilosis*, *C. krusei*, and *C. glabrata*, along with *C. albicans*, now account for over **90% of candidemia cases**, posing a significant threat to the health of **hospitalized and immunocompromised patients**.

Aim of the Work

- To investigate the **prevalence of *Candida*** as a causative agent of **candidemia** and **bloodstream infections**.
- To detect **biofilm formation** using the **Tissue Culture Plate (TCP) method**.
- To assess the **prevalence of antifungal resistance**, biofilm formation, and the presence of **virulence genes** (such as *ALS1* and *HWP1*) in clinical isolates of *Candida albicans* obtained from **immunocompromised patients**.
- To detect **biofilm-associated virulence genes** by **Polymerase Chain Reaction (PCR)**.
- To perform **antifungal susceptibility testing** for the isolated *Candida albicans* strains.
- To specifically detect the **ALS1 and HWP1 genes**, which are involved in biofilm formation in *Candida albicans*.

Materials and Methods

1 Sample Collection

Blood and urine samples were collected aseptically from each patient under strict antiseptic conditions.

2 Culture Media

- **Sabouraud Dextrose Agar (SDA)** – used for the primary isolation of fungi.
- **Chromogenic Candida Agar (CCA)** (TmMedia, India) – utilized for presumptive identification of *Candida* species based on colony color.

3 Identification of *Candida* Species

Isolated colonies were subjected to further identification using the following methods:

- **Gram Staining:** Revealed violet-colored budding yeast cells.
- **Sugar Fermentation Test:** To differentiate species based on carbohydrate utilization.
- **Urease Test:** Used to distinguish urease-producing strains.
- **Germ Tube Test:** For the rapid presumptive identification of *Candida albicans*.

4 Detection of Biofilm Formation

- Congo Red Agar (CRA) Method
- Tissue Culture Plate (TCP) Method

5 Molecular Identification of Biofilm-Related Genes (Conventional PCR)

Table 2. Primers used for detection of biofilm formation genes in *C. albicans*.

Primer	Sequences (5' to 3')	Amplicon (bp)	References
<i>ALS1-F</i>	5'-GAC TAG TGA ACC AAC AAA TAC CAG A -3'	318	[13]
<i>ALS1-R</i>	5'-CCA GAA GAA ACA GCA GGT GA -3'		
<i>HWP1-F</i>	5'-TCAGTTCCACTCATGCAACCA-3'	99	[14]
<i>HWP1-R</i>	5'-AGCACCGAAAGTCAATCTCATGT-3'		

6 Antifungal Susceptibility Testing by

- culture media : Mueller-Hinton Agar (Oxoid)
- Antifungal Susceptibility Discs (Azoles , Polyenes and Echinocandins)

Ethics statement

Patients will be informed on the study details and their written consent will be obtained and documented. The study protocol and consent procedure will be approved by ethics committee of the Faculty of Medicine, Assuit University.

References

- 1 Gow NA, van de Veerdonk FL, Brown AJ, Netea MG. Candida albicans morphogenesis and host defence: Discriminating invasion from colonization. Nat Rev Microbiol 2011; 10(2): 112-22. <http://dx.doi.org/10.1038/nrmicro2711> [PMID: 22158429]
- 2 Karacaer Z, Oncul O, Turhan V, Gorennek L, Ozyurt M. A surveillance of nosocomial candida infections: Epidemiology and influences on mortality in intensive care units. Pan Afr Med J 2014; 19: 398-4960. <http://dx.doi.org/10.11604/pamj.2014.19.398.4960> [PMID: 25995794]
- 3 Pfaller MA, Diekema DJ. Epidemiology of invasive mycoses in North America. Crit Rev Microbiol 2010; 36(1): 1-53. <http://dx.doi.org/10.3109/10408410903241444> [PMID: 20088682]
- 4 Williams DW, Jordan RP, Wei XQ, et al. Interactions of Candida albicans with host epithelial surfaces. J Oral Microbiol 2013; 5: 5. <http://dx.doi.org/10.3402/jom.v5i0.22434> [PMID: 24155995]
- 5 Spellberg B. Novel insights into disseminated candidiasis: Pathogenesis research and clinical experience converge. PLoS Pathog 2008; 4(2)e38<http://dx.doi.org/10.1371/journal.ppat.0040038> [PMID: 18282100]
- 6 Van De Veerdonk FL, Kullberg BJ, Netea MG. Pathogenesis Of Invasive Candidiasis. Curr Opin Crit Care 2010; 16(5): 453-9. [Http://Dx.Doi.Org/10.1097/Mcc.0b013e32833e046e](http://Dx.Doi.Org/10.1097/Mcc.0b013e32833e046e) [Pmid:20711079]
- 7 Kaur H, Chakrabarti A. Strategies To Reduce Mortality In Adult And Neonatal Candidemia In Developing Countries. J Fungi (Basel) 2017; 3(3): 41. [Http://Dx.Doi.Org/10.3390/Jof3030041](http://Dx.Doi.Org/10.3390/Jof3030041) [PMID: 29371558]
- 8 Galocha M, Pais P, Cavalheiro M, Pereira D, Viana R, Teixeira MC. Divergent Approaches To Virulence In C. Albicans And C. Glabrata: Two Sides Of The Same Coin. Int J Mol Sci 2019; 20(9): 2345. [Http://Dx.Doi.Org/10.3390/Ijms20092345](http://Dx.Doi.Org/10.3390/Ijms20092345) [PMID: 31083555]
- 9 Sundstrom P, Balish E, Allen CM. Essential Role Of The Candida Albicans Transglutaminase Substrate, Hyphal Wall Protein 1, In Lethal Oroesophageal Candidiasis In Immunodeficient Mice. J Infect Dis 2002; 185(4): 521-30. [Http://Dx.Doi.Org/10.1086/338836](http://Dx.Doi.Org/10.1086/338836) [Pmid: 11865405]
- 10 Fu Y, Ibrahim AS, Sheppard DC, et al. Candida albicans Als1p: An adhesin that is a downstream effector of the EFG1 filamentation pathway. Mol Microbiol 2002; 44(1): 61-72. <http://dx.doi.org/10.1046/j.1365-2958.2002.02873.x> [PMID: 11967069]



Thank you